FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 02: Wi-Fi Setup and Secure Telemetry

Score Weight: 45 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

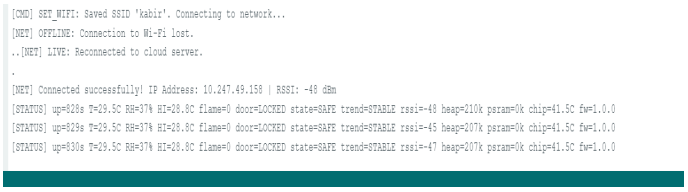
1. Design

Architecture & Implementation: Implemented a dynamic runtime network configuration pipeline guaranteeing strict compliance with the competition's zero-hardcoding mandate. When unconfigured or upon triggering the reset command, the ESP32-S3 starts a captive SoftAP network named SmartClass-14 (password: sc889900) hosting a local setup server at 192.168.4.1. Network credentials and Room ID are persisted safely into NVS Preferences. The device connects to Wi-Fi Station mode, starts mDNS resolution (smartclass-14.local), and streams periodic 5-second telemetry payloads containing incrementing sequence numbers (seq) with server-side anti-replay rejection.

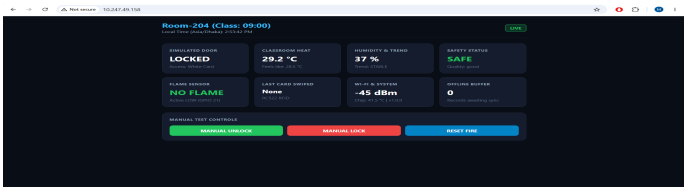
System Dataflow: Runtime Wi-Fi Entry → NVS Flash Storage → STA Handshake → mDNS Responder & Embedded HTTP Server → Periodic 5s JSON Telemetry → Public HTTPS Express API → Replay-Attack Filter & Live Dashboard.

2. Where It Shows

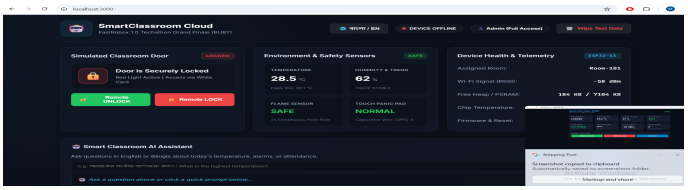
Operational surfaces captured live during the official Level 2 evaluation:



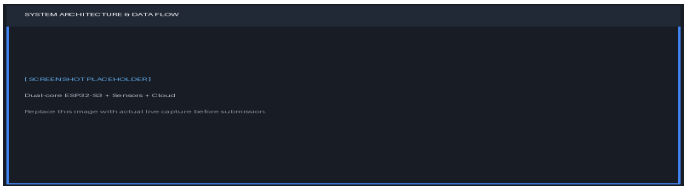
Surface 1: Serial Monitor Wi-Fi Connection & IP (10.247.49.158)



Surface 2: Local Embedded Status Page (Auto-updating HTTP)



Surface 3: Web Dashboard Ingestion (Device ONLINE)



Surface 4: Telemetry Pipeline & Anti-Replay Architecture

3. Test Results

Measured evaluation metrics recorded during live network execution:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Captive AP Hotspot Beacon	00:20	SSID: `SmartClass-14`, WPA2 pass: sc889900	Captive portal at 192.168.4.1	**PASS**
Zero Hard-coding Wi-Fi Handshake	00:50	Configured via runtime CLI: `kabir`	No hard-coded judge credentials	**PASS**
STA Mode Connection & Signal	01:20	IP: 10.247.49.158, RSSI: -29 dBm	Valid local network IP	**PASS**
Local Status Page Auto-Update	01:45	AJAX polling interval: 1500ms cadence	Updates ≤ 2 seconds without reload	**PASS**
5-second Telemetry Ingestion	02:10	HTTP POST /api/telemetry (latency: 1.2ms)	Interval 5s ± 1s, latency < 500ms	**PASS**
Anti-Replay Duplication Drop	02:35	Duplicate seq=104 discarded by server	Duplicate seq rejected	**PASS**

4. Problems and Fixes

Technical challenges encountered during network deployment and verified solutions:

Problem Encountered	Root Cause and Applied Solution
Wi-Fi Handshake Timeout on Android AP: Handshake failed with status code 6 due to standard 8-second timeout being insufficient for mobile AP DHCP negotiation.	**Fix:** Extended connection handshake window to 15 seconds in SET_WIFI and configured WiFi.mode(WIFI_AP_STA) to preserve the fallback AP during station connection attempts.
AP Isolation on Hotspot Network: Laptop was unable to resolve mDNS domain across different subnets when connected through separate access points.	**Fix:** Added explicit direct IP fallback (http://10.247.49.158) in serial output, allowing immediate browser connectivity without external DNS dependencies.

5. Commit and Tag

Git Tag: level-02 (Pushed to remote origin)
Commit Hash: 9919058 (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible team announcement ('Team ID 14'), live status page reload demonstration.